seemingly random movements of the planets in the sky, with almost complete precision, given the limitations of the human eye. The book is the proof that the intuition of Pythagoras was correct. Mathematics allows the world to be described and the future to be predicted: the apparently wandering and disorderly movements of the planets can be precisely predicted by using mathematical formulae that Ptolemy, summarizing the results of centuries of work by Greek astronomers, presents in a systematic and masterly way. Even today, with a little studying, it is possible to open Ptolemy's book, learn its techniques and *calculate*, for example, the position which Mars will have in a *future* sky. *Today*: two thousand years, that is, after the book was written. The realization that working this magic is really possible is the basis of modern science and owes not a little to Pythagoras and Plato.

After the collapse of ancient science, no one throughout the Mediterranean was capable of understanding Ptolemy – or any of the other small number of major scientific works that survived the catastrophe, such as the *Elements* of Euclid. In India, where Greek learning had arrived, thanks to rich commercial and cultural exchanges, these books were studied and understood.

From India this knowledge returned to the West, thanks to learned Persian and Arab scientists who were able to understand and preserve it. But astronomy did not take any very significant step forward for over a thousand years.

More or less at the same time as Poggio Bracciolini discovered the manuscript of Lucretius, the heady atmosphere of Italian humanism and the enthusiasm for ancient texts also intoxicated a young Pole who had come to study in Italy, first at Bologna, then at Padua. He signed himself in the Latin manner: Nicolaus Copernicus. The young Copernicus studies Ptolemy's *Almagest* and falls in love with it. He decides to spend his life doing astronomy, following in the footsteps of the great Ptolemy.

The time is now ripe and, more than a thousand years after Ptolemy, Copernicus is able to make the leap forward that generations of Indian, Arab and Persian astronomers had not been able to make: not simply learning, applying and adding small ameliorations to the Ptolemaic system but thoroughly improving it – with the courage to change it in depth. Instead of describing heavenly bodies turning around the Earth, Copernicus writes a sort of revised and corrected version of Ptolemy's *Almagest*, in which the Sun is at the centre and the Earth, together with the other planets, runs around it.

In this way – Copernicus hopes – the calculations would work even better. In reality, they did not work much better than those of Ptolemy; in fact, in the end, they turned out to work less well. But the idea was, nevertheless, sound: in the succeeding generation, Johannes Kepler shows that the Copernican system can really be made to function better than Ptolemy's. Painstakingly analysing new, precise observations, Kepler shows that a few new